

Machine Learning-Lecture 9

P1.py

```
C:\ml\L9>python p1.py
```

	DN	Weather	Car	Result
0	1	sunny	working	go-out
1	2	rainy	broken	go-out
2	3	sunny	working	go-out
3	4	sunny	working	go-out
4	5	sunny	working	go-out
5	6	rainy	broken	stay-home
6	7	rainy	broken	stay-home
7	8	sunny	working	stay-home
8	9	sunny	broken	stay-home
9	10	rainy	broken	stay-home

	Weather_rainy	Weather_sunny	Car_broken	Car_working
0	False	True	False	True
1	True	False	True	False
2	False	True	False	True
3	False	True	False	True
4	False	True	False	True
5	True	False	True	False
6	True	False	True	False
7	False	True	False	True
8	False	True	True	False
9	True	False	True	False

weather: 1 for rainy and 2 for sunny 2

car:1 for broken and 2 for working 2

['go-out']

['go-out' 'stay-home']

[[0.94553707 0.05446293]]

P2.py

```
C:\ml\L9>python p2.py
```

```
free win offer buy spam
```

```
0  1  1  0  1 Yes
1  0  0  0  0 No
2  1  0  1  1 Yes
3  0  1  0  0 No
4  1  1  1  1 Yes
5  1  0  1  1 Yes
6  0  1  0  0 No
7  0  0  0  0 No
8  1  1  1  0 Yes
9  0  0  1  1 No
10 1  0  0  0 No
11 1  1  1  1 Yes
12 0  1  0  0 No
13 0  0  1  1 No
14 1  1  1  1 Yes
15 0  0  0  0 No
16 1  0  1  0 Yes
17 0  1  0  0 No
18 1  1  1  1 Yes
19 1  1  1  1 Yes
```

```
free ->0 if not present and 1 if present 1
```

```
win->0 if not present and 1 if present 1
```

```
offer->0 if not present and 1 if present 0
```

```
buy->0 if not present and 1 if present 1
```

```
['Yes']
```

P3.py

```
C:\ml\L9>python p3.py
```

```
enter number of hours studied 6
```

```
enter number of hours slept 3
```

```
enter stress level 2
```

```
['No']
```

```
['No' 'Yes']
```

```
[[0.92097115 0.07902885]]
```

P4.py

```
C:\ml\L9>python p4.py
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
..	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

```
[400 rows x 5 columns]
```

```
User ID      0
```

```
Gender       0
```

```
Age          0
```

```
EstimatedSalary  0
```

Purchased 0

dtype: int64

	Age	EstimatedSalary	Gender_Female	Gender_Male
0	19	19000	False	True
1	35	20000	False	True
2	26	43000	True	False
3	27	57000	True	False
4	19	76000	False	True
..	...	...	...	...
395	46	41000	True	False
396	51	23000	False	True
397	50	20000	True	False
398	36	33000	False	True
399	49	36000	True	False

[400 rows x 4 columns]

Purchased

0 257

1 143

Name: count, dtype: int64

	precision	recall	f1-score	support
0	0.92	0.94	0.93	64
1	0.89	0.86	0.87	36
accuracy			0.91	100
macro avg	0.90	0.90	0.90	100
weighted avg	0.91	0.91	0.91	100

enter age 19

enter salary 19000

Gender--> 1 female and 2 male 2

[0]

[[0.98400903 0.01599097]]